

# Devana Perupurayil

San Diego, CA | [devanaperupurayil@gmail.com](mailto:devanaperupurayil@gmail.com) | (619) 953-7200 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

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## EDUCATION

**University of California, San Diego, CA**

**CGPA: 3.89/4**

*Master of Science in Data Science*

Graduation: June 2026

Relevant Coursework: Machine Learning,, Optimization, Algorithms for Data Science, Data Management, Statistical Modeling, NLP

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## SKILLS

- Programming: Python, SQL, R, Java
  - Machine Learning: Time Series Forecasting, Statistical Modeling, NLP, Transformer Models, Bayesian Inference, Classification, Clustering
  - Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, XGBoost
  - Data & Visualization: Snowflake, PostgreSQL, Power BI, NetSuite, Pandas
  - Tools/Platforms: Jenkins, Kubernetes, Git
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## WORK EXPERIENCE

**MADE BY GATHER, NEW YORK, NY (REMOTE)**

*Aug 25 – June 26*

**Data Analyst Intern**

- Developed retailer-level weekly sales forecasting models using Holt-Winters, ARIMA, Random Forest, and XGBoost in Python, achieving >70% forecasting accuracy across retailers using limited historical datasets.
- Built a seasonality-aware forecasting framework combining statistical and machine learning approaches, improving forecast accuracy by 50% and reducing forecasting errors by 30%.
- Automated the end-to-end weekly forecasting pipeline using Jenkins, reducing manual reporting effort from 4 hours to 1 hour.
- Reverse-engineered NetSuite sales and finance reporting pipelines into Snowflake SQL, resolving aggregation inconsistencies and improving reporting accuracy.
- Designed Power BI dashboards for daily sales analytics and executive reporting, replacing manual Excel-based workflows.

**ORACLE FINANCIAL SERVICES SOFTWARE LTD., Bangalore, India**

*July 22 – Aug 24*

**Associate Software Developer**

- Developed and tested backend services and REST APIs using Java and Spring Boot, improving payment processing reliability and reducing transaction errors by 20%.
  - Maintained and deployed 40+ services using CI/CD pipelines, shell scripting, and Tomcat server infrastructure to improve system reliability and debugging efficiency.
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## ACADEMIC PROJECTS

**Motherese: Prosody-Aware Language Modeling using Whisper | PyTorch, Transformers**

*Apr 25 – June 25*

- Developed a prosody-enhanced variant of OpenAI's Whisper model using PyTorch and Transformer architectures to study syntactic learning in speech.
- Engineered custom embeddings integrating pitch, duration, and suprasegmental speech features, improving dependency parsing F1 scores by 35%.
- Built automated preprocessing pipelines for 205+ hours of speech data using Montreal Forced Aligner and Kubernetes-based GPU training workflows.
- Evaluated model performance on CHILDES datasets, achieving up to 66% F1 score improvements on complex long-range dependency tasks.

**MJD Data Release Analysis – Neutrinoless Double Beta Decay Detection | Python**

*Mar 25*

- Applied Bayesian statistical modeling and Markov Chain Monte Carlo (MCMC) sampling to analyze Majorana Demonstrator detector signals for potential neutrinoless double beta decay events.
- Developed classification models and ROC-based evaluation pipelines, achieving an AUC of 0.86 for signal-background separation.
- Performed Gaussian signal modeling and energy spectrum analysis to estimate decay signal probabilities and improve event detection accuracy.